

# sentry edu 无线远程连接

< sentry edu, Edit connection

NOMACHINE



Address  
Name, host, port and protocol



Configuration  
Authentication and multimedia



Info  
Model, OS and product version

## Machine address

Connect

Remove



sentry edu

Direct connection over the Internet.

Give a name and save the settings for your connection.

Name sentry edu

Insert a host name, IP address or Machine Id.

Host 10.176.122.118

Port 4000

Protocol NX

☒ Always accept the host verification key provided by the remote host

The verification key is used to ensure you connect to the server you want.



Reset saved preferences and password

# sentry wire 有线连接

[<](#) sentry wire, Edit connection

NOMACHINE

 Address  
Name, host, port and protocol

 Configuration  
Authentication and multimedia

 Info  
Model, OS and product version

 sentry wire  
Direct connection over the Internet.

Give a name and save the settings for your connection.

Name

Insert a host name, IP address or Machine Id.

Host  Port  Protocol

☒ Always accept the host verification key provided by the remote host  
The verification key is used to ensure you connect to the server you want.

 Reset saved preferences and password

## 三、 标准操作程序 (SOP): 系统一键复现指南

为了保证后续复现 100% 成功，每次开机后，请严格开 4 个独立的终端，按以下顺序执行操作：

### 终端 1：唤醒感知驱动层 (满血启动相机)

**目的：** 开启双目红外、IMU 插值同步、禁用多余深度流压缩，直出 3D 稠密点云。

Bash

```
1 # 1. 进门与挂载
2 cd ~/d455_ws
3 source start_env.sh
4
5 # 2. 满血点火
6 ros2 launch realsense2_camera rs_launch.py \
7     enable_infra1:=true \
8     enable_infra2:=true \
9     enable_gyro:=true \
10    enable_accel:=true \
11    unite_imu_method:=1 \
12    enable_sync:=true \
13    depth_module.profile:=848x480x30 \
14    pointcloud.enable:=true
```

## 终端 2：唤醒视觉里程计层 (启动 VINS 大脑)

目的：启动核心解算引擎。

Bash

```
1 # 1. 进门与挂载
2 cd ~/vins_ws
3 source start_env.sh
4
5 # 2. 启动解算节点
6 ros2 run vins vins_node /home/soyo/vins_ws/src/VINS-Fusion-
  ROS2/config/realSense_d455/d455_stereo_imu_config.yaml
```

## 终端 3：搭建 TF 坐标桥 (连接两个平行宇宙)

目的：解决 VINS 不发 TF 的历史遗留 Bug，将 `world` 与相机的物理坐标系锁定。

Bash

```
1 # 1. 进门与挂载基础 ROS
2 cd ~/vins_ws
3 source /opt/ros/humble/setup.bash
4
5 # 2. 运行桥接节点
6 ./tf_bridge.py
```

## 终端 4：唤醒全局监视面板 (启动 RViz)

目的：在 NoMachine 桌面中加载我们捏好的 100x100 战车专用面板。

Bash

```
1 # 挂载基础 ROS 并启动一键脚本
2 source /opt/ros/humble/setup.bash
3 ros2 launch ~/vins_ws/start_viz.launch.py
```

## 四、 极其关键：VINS 的“唤醒舞蹈” (初始化规范)

由于视觉惯性里程计 (VIO) 存在尺度不确定性，**严禁开局直接把相机拿起来乱转！** 启动上述 4 个终端后，必须执行以下物理操作，否则在 RViz 中机器人会瞬间“遁地”或“飞天”：

- 开局静止**：启动指令前，相机必须平放在桌面上，正前方必须有丰富的视觉纹理（不要对着纯色墙壁）。
- 纯平移唤醒**：端起相机，**保持镜头朝向不变**，缓慢向左平移 20cm，再向右平移 20cm。
- 上下视差**：再缓慢向上提 20cm，向下压 20cm。
- 确认收敛**：观察 RViz 屏幕，如果代表相机的 XYZ 坐标轴和绿色轨迹稳定跟随你的手部移动，说明 IMU 的尺度和重力向量已经成功标定，此时可以随意走动。



```
1/5 ~ + [T] [C] Tilix: soyo@soyo-mygo: ~/d455_ws [Q] [≡] - □ ×
1: soyo@soyo-mygo: ~/d455_ws □ ×
[realsense2_camera_node-1] [WARN] [1780131034.322011819] [camera.camera]: XXX Hardware Notification:Frames didn't arrived within 5 seconds,1.78013e+12,Warn,Frames Timeout
[realsense2_camera_node-1] 30/05 16:50:34,396 WARNING [140229891802688] (backend-v4l2.cpp:1858) Frames didn't arrived within 5 seconds
[realsense2_camera_node-1] [WARN] [1780131034.397042541] [camera.camera]: XXX Hardware Notification:Frames didn't arrived within 5 seconds,1.78013e+12,Warn,Frames Timeout
[realsense2_camera_node-1] 30/05 16:50:39,314 WARNING [140229910443584] (backend-v4l2.cpp:1858) Frames didn't arrived within 5 seconds
[realsense2_camera_node-1] [WARN] [1780131039.314715667] [camera.camera]: XXX Hardware Notification:Frames didn't arrived within 5 seconds,1.78013e+12,Warn,Frames Timeout
[realsense2_camera_node-1] 30/05 16:50:39,327 WARNING [140229902050880] (backend-v4l2.cpp:1858) Frames didn't arrived within 5 seconds
[realsense2_camera_node-1] [WARN] [1780131039.327086698] [camera.camera]: XXX Hardware Notification:Frames didn't arrived within 5 seconds,1.78013e+12,Warn,Frames Timeout
[realsense2_camera_node-1] 30/05 16:50:39,401 WARNING [140229891802688] (backend-v4l2.cpp:1858) Frames didn't arrived within 5 seconds
[realsense2_camera_node-1] [WARN] [1780131039.402011089] [camera.camera]: XXX Hardware Notification:Frames didn't arrived within 5 seconds,1.78013e+12,Warn,Frames Timeout

2/5 ~ + [T] [C] Tilix: soyo@soyo-mygo: ~/vins_ws [Q] [≡] - □ ×
1: soyo@soyo-mygo: ~/vins_ws □ ×
[WARN] [1780130129.713549859] []: Optimize extrinsic param around initial guess
camera number 2
[INFO] [1780130129.713834289] []: Unsynchronized sensors, online estimate time offset, initial td: 0.000000
[INFO] [1780130129.713854041] []: ROW: 480 COL: 848
extrinsic cam 0
1 0 0
0 1 0
0 0 1
-0.0302 0.0074 0.016
extrinsic cam 1
1 0 0
0 1 0
0 0 1
-0.1252 0.0074 0.016
set g 0 0 9.805
[INFO] [1780130129.713942541] []: reading parameter of camera /home/soyo/vins_ws/src/VINS-Fusion-ROS2/config/realsense_d455/left.yaml
[INFO] [1780130129.714224374] []: reading parameter of camera /home/soyo/vins_ws/src/VINS-Fusion-ROS2/config/realsense_d455/right.yaml
MULTIPLE_THREAD is 1
[WARN] [1780130129.714475466] []: waiting for image and imu...
```

```
3/5 v + [T] [C] Tilix: soyo@soyo-mygo: ~/vins_ws [Q] [≡] [-] [□] [X]
1: soyo@soyo-mygo: ~/vins_ws v [□] [X]
[setupvars.sh] OpenVINO environment initialized
soyo@soyo-mygo:~/vins_ws$ ^[[200~cd ~/vins_ws
cd: command not found
soyo@soyo-mygo:~/vins_ws$ source /opt/ros/humble/setup.bash~cd ~/vins_ws
source /opt/ros/humble/setup.bash
bash: /opt/ros/humble/setup.bash~cd: No such file or directory
soyo@soyo-mygo:~/vins_ws$
soyo@soyo-mygo:~/vins_ws$ cd ~/vins_ws
source /opt/ros/humble/setup.bash
soyo@soyo-mygo:~/vins_ws$ ./tf_bridge.py
[INFO] [1780130155.966638192] [vins_tf_bridge]: 🚢 VINS TF Bridge 已启动! 正在强行桥接 world 和 camera_imu_frame...
```

```
4/5 v + [T] [C] Tilix: soyo@soyo-mygo: ~/vins_ws [Q] [≡] [-] [□] [X]
1: soyo@soyo-mygo: ~/vins_ws v [□] [X]
[setupvars.sh] OpenVINO environment initialized
soyo@soyo-mygo:~/vins_ws$ source /opt/ros/humble/setup.bash
soyo@soyo-mygo:~/vins_ws$ ros2 launch ~/vins_ws/start_viz.launch.py
[INFO] [launch]: All log files can be found below /home/soyo/.ros/log/2026-05-30-16-36-17-416290-soyo-mygo-15608
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [rviz2-1]: process started with pid [15609]
[rviz2-1] Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway.
[rviz2-1] [INFO] [1780130177.749620050] [rviz2]: Stereo is NOT SUPPORTED
[rviz2-1] [INFO] [1780130177.749675446] [rviz2]: OpenGL version: 3.3 (GLSL 3.3)
[rviz2-1] [INFO] [1780130177.766454569] [rviz2]: Stereo is NOT SUPPORTED
```

```
5/6 v + [T] [C] Tilix: soyo@soyo-mygo: ~/vins_ws [Q] [≡] [-] [□] [X]
1: soyo@soyo-mygo: ~/vins_ws v [A] [□] [X]
/camera/extrinsics/depth_to_color
/camera/extrinsics/depth_to_depth
/camera/extrinsics/depth_to_gyro
/camera/extrinsics/depth_to_infra1
/camera/gyro/imu_info
/camera/gyro/metadata
/camera/gyro/sample
/camera/imu
/camera/infra1/camera_info
/camera/infra1/image_rect_raw
/camera/infra1/image_rect_raw/compressed
/camera/infra1/image_rect_raw/compressedDepth
/camera/infra1/image_rect_raw/theora
/camera/infra1/metadata
/camera/infra2/image_rect_raw
/camera_pose
/camera_pose_visual
/chassis_command
/clicked_point
/cmd_spin
/cmd_vel
```